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Department of Mechanical Engineering
RV College of Engineering®, Bengaluru – 560059

ELEMENTS OF MECHANICAL ENGINEERING

UNIT-II

FUNDAMENTALS OF MACHINE TOOLS AND OPERATIONS

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UNIT-II

Lathe and Lathe operations: Classification, specifications of a lathe. Lathe operations (Turning, Taper Turning, drilling, boring, knurling, and thread cutting). Introduction to CNC Machines.

Joining processes & Non-destructive testing

Introduction to metal joining process-permanent & temporary joints, Soldering & welding, types and applications, accessories consumables and safety, Welding defects and causes, Non-Destructive testing: Liquid penetrate testing, Magnetic particle testing, Ultrasonic testing, Eddy current testing.

MACHINE TOOL AND MACHINING

- **Machining** is the process of removing the excess material from the work piece in the form of chips, by forcing a cutting tool with one or more cutting edges to arrive the required shape and size.
- **Machine tool** is a power driven machine to perform machining. Generally, a machine tools shall produce either cylindrical or flat surfaces on majority.

A machine tool performs three major functions

- It rigidly supports the work piece and the cutting tool.
- It provides relative motion between the work piece and the cutting tool
- It provides a range of speeds & feeds.

Lathe machine Classifications :

The most commonly used machine tools are;

- Lathe
- Grinding machine
- Drilling Machine
- Shaping Machine
- Planning Machine
- Milling Machine

LATHE MACHINE

Working Principle:

- Lathe is a machine tool which holds the work in a holding device and rotating it against a suitable cutting tool to remove the excess metal from the work.
- If the tool is moved parallel to the axis of rotation of the work, a cylindrical surface is produced.
- If the tool is moved perpendicular to the axis of rotation of the work, a flat surface is produced.

Lathe machine classifications:

Lathes are classified as;

(i) Bench Lathe

(ii) Speed lathe

(iii) Engine Lathe

(iv) Tool room lathe

(v) Capstan & Turret lathe

(vi) Automatic lathe

(vii) Special purpose lathe

BENCH LATHE



SPEED LATHE



ENGINE LATHE (OR) CENTER LATHE



TOOL ROOM LATHE



CAPSTAN AND TURRET LATHE



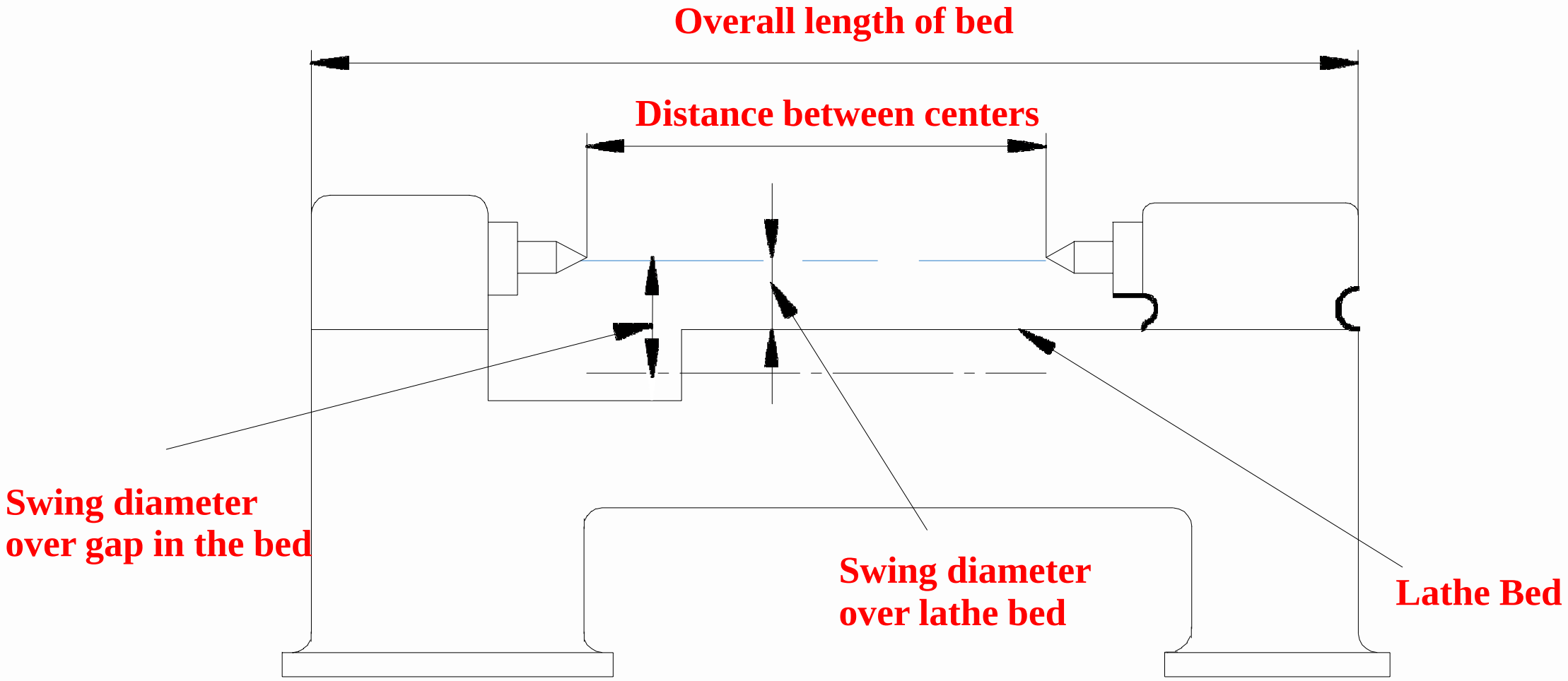
AUTOMATIC LATHE



SPECIAL PURPOSE LATHE



LATHE SPECIFICATIONS :



LATHE SPECIFICATIONS :

The size of a lathe is specified by the following:

1. ***Maximum diameter of the work piece that can be revolved over the lathe bed.***

It is also called as height of centers above the lathe bed Or Swing diameter over lathe bed.

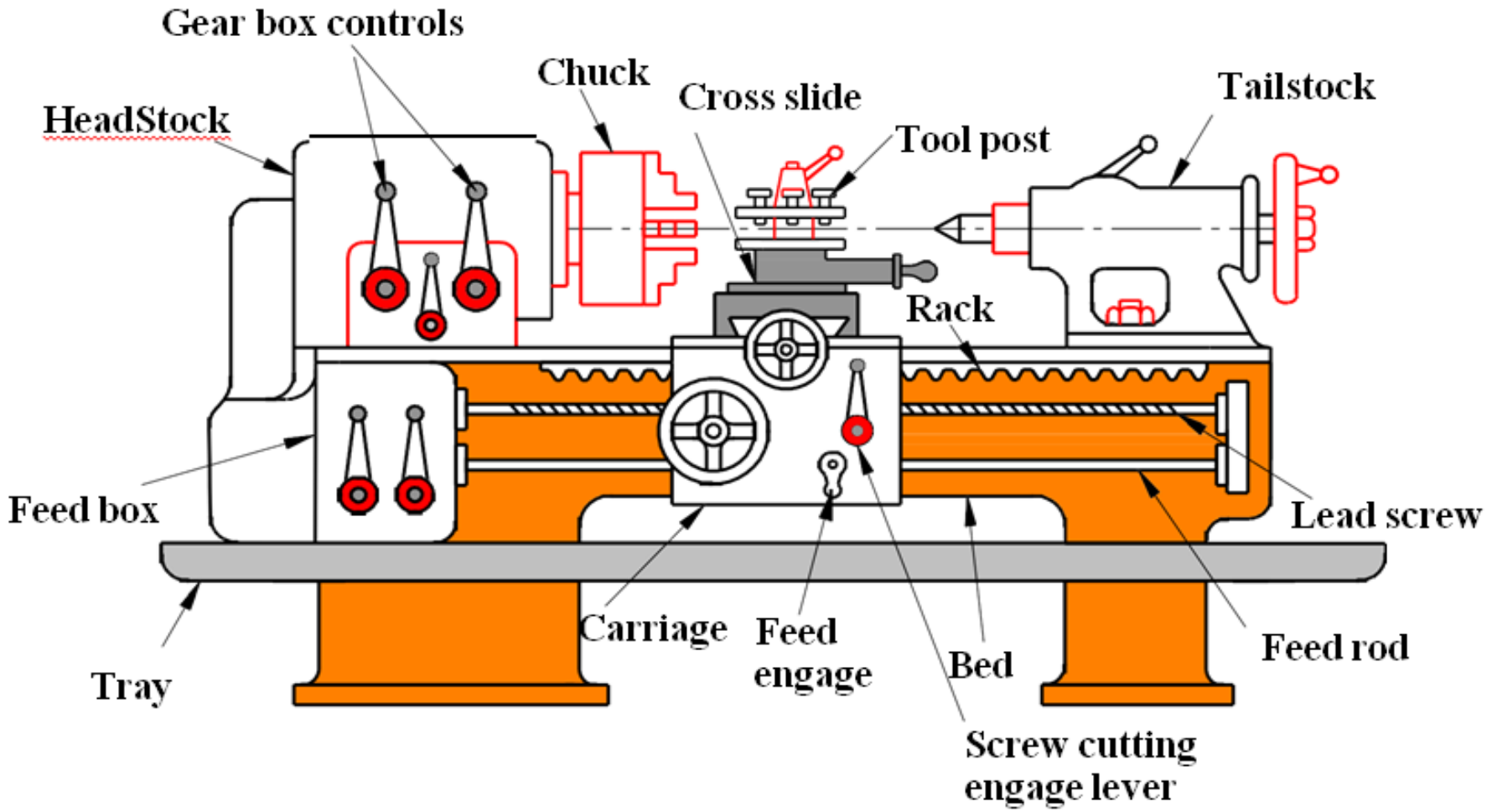
2. ***Maximum diameter and width of the work piece,*** that can swing when the lathe has a gap bed.

3. ***Maximum length of work piece*** ,that can be mounted between centers.

4. ***Overall length of bed:*** It is the total length of the lathe itself.

5. ***Maximum number of speeds & feeds available.***

PARTS OF LATHE



Parts of a Lathe

Major parts included in lathe construction :

The major parts included in lathe construction :

1. Bed
2. Headstock
3. Tailstock
4. Carriage
5. Feed rod
6. Lead screw

Major parts included in lathe construction :

1. Bed:

- The lathe bed forms the support for all other parts.
- At the top of the bed, precision machined guide ways (inner & outer) are present.
- The headstock & tailstock are mounted on the inner ways which keep them perfectly aligned with each other.
- The outer ways guide the movement of carriage assembly.
- Bed is usually made of grey cast iron because of its good damping property and low frictional resistance.



Major parts included in lathe construction :

2. Headstock :

- Headstock is a box like casting mounted at the left end of the machine.
- It consists of a spindle and gear box to rotates the spindle at different speeds.
- The spindle has external threads to receive a work holding device such as a chuck or face plate.
- The gear box distributes power to the feed rod and leads screw for turning or thread cutting operation.





Major parts included in lathe construction :

3. Tailstock :

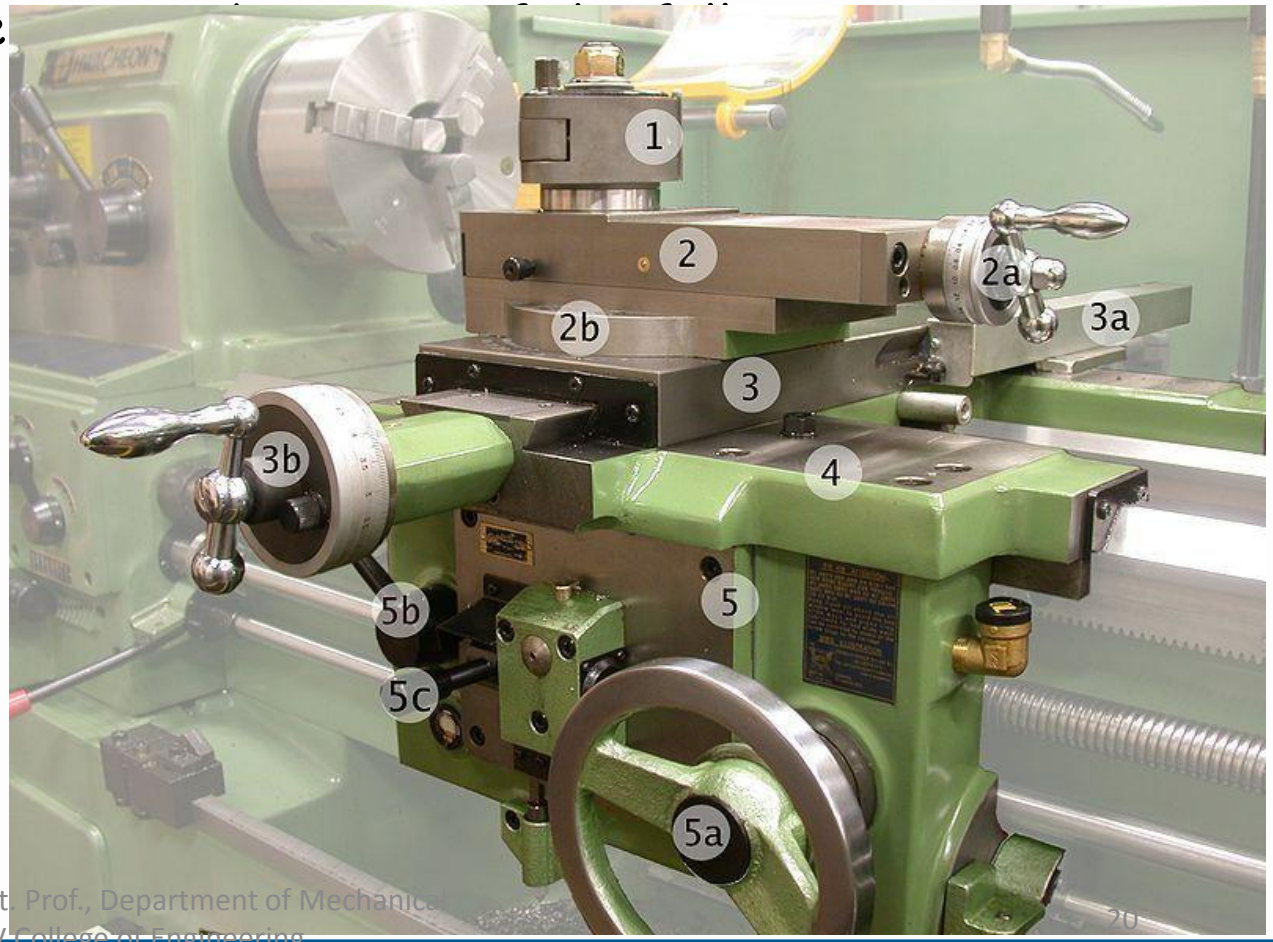
- Tailstock is mounted at the right end of the machine.
- It may be slides on the guide ways to support different lengths of work pieces between centers and clamped to bed at any desired location.
- It carries a dead center or revolving center.
- Tailstock can be offset for taper turning. It can also carry tools like drill bit, reamer, tap, etc.



Major parts included in lathe construction :

4. Carriage Assembly

- it supports and feeds the cutting tool against the work during operation of the lathe.
- The carriage slides along the outer guide
  1. Saddle (Part no. 03A)
  2. Cross slide (Part no. 03)
  3. Compound slide (Part no. 02,02B)
  4. Tool post (Part no. 01)
  5. Apron (Part no. 05)



Major parts included in lathe construction :

Saddle :

It is a H-shaped casting that slides over the outer guide ways and serves as a base for the cross slide.

Cross slide:

It is mounted on the saddle and enables the movement of the cutting tool laterally across the lathe bed by means of a cross feed hand wheel.

Compound rest:

It is mounted on the top of the cross slide and supports the tool post. It can be swiveled to any angle to facilitate taper turning & threading operations.

Tool post:

It is mounted on the top of the compound rest and carries the cutting tool

Apron:

The portion of the carriage that extends in front of the lathe is called an apron. It houses feed mechanisms. It contains pinion which engages with the rack for longitudinal feed.

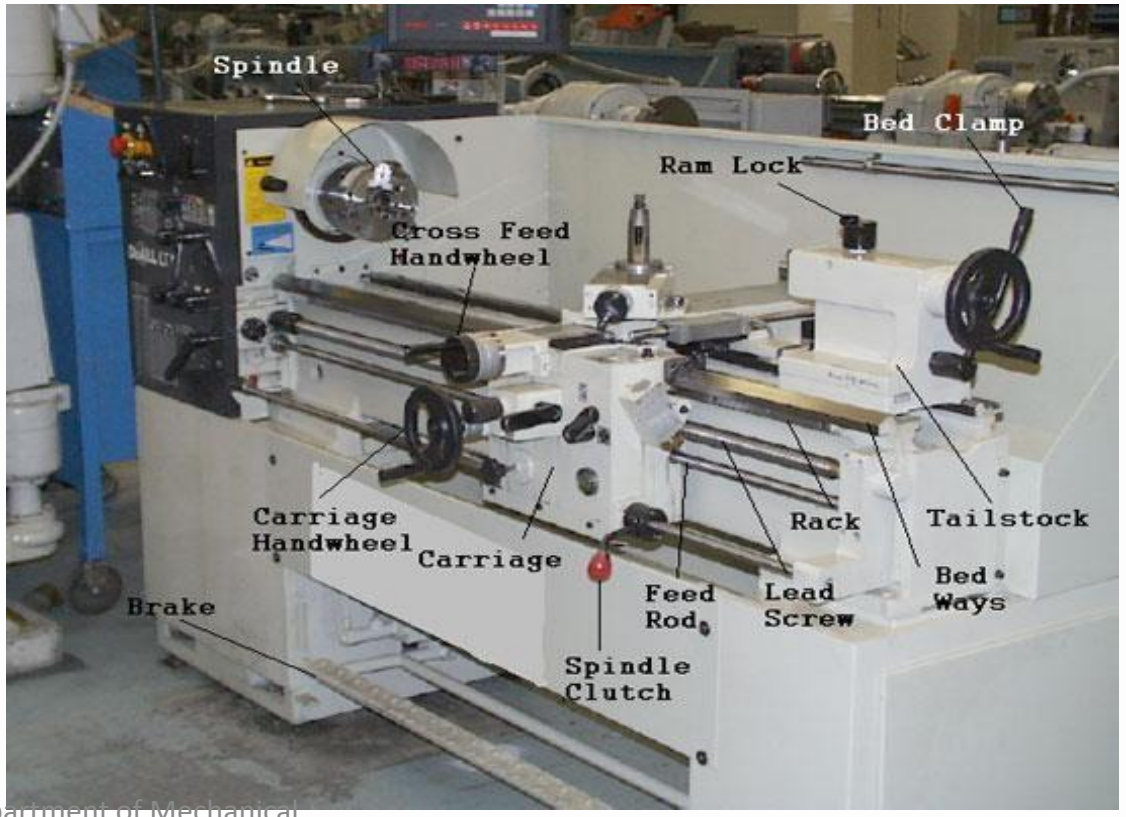
Major parts included in lathe construction :

5. Feed Rod:

- The feed rod is a long shaft used to drive the apron mechanism for cross & longitudinal power feed during turning, boring & facing operations. The feed rod is powered by a set of gears from the headstock.

6. Lead Screw:

- It is a long threaded shaft geared to the headstock. Closing a split nut around the lead screw engages it with the carriage. It used only for thread cutting operations.



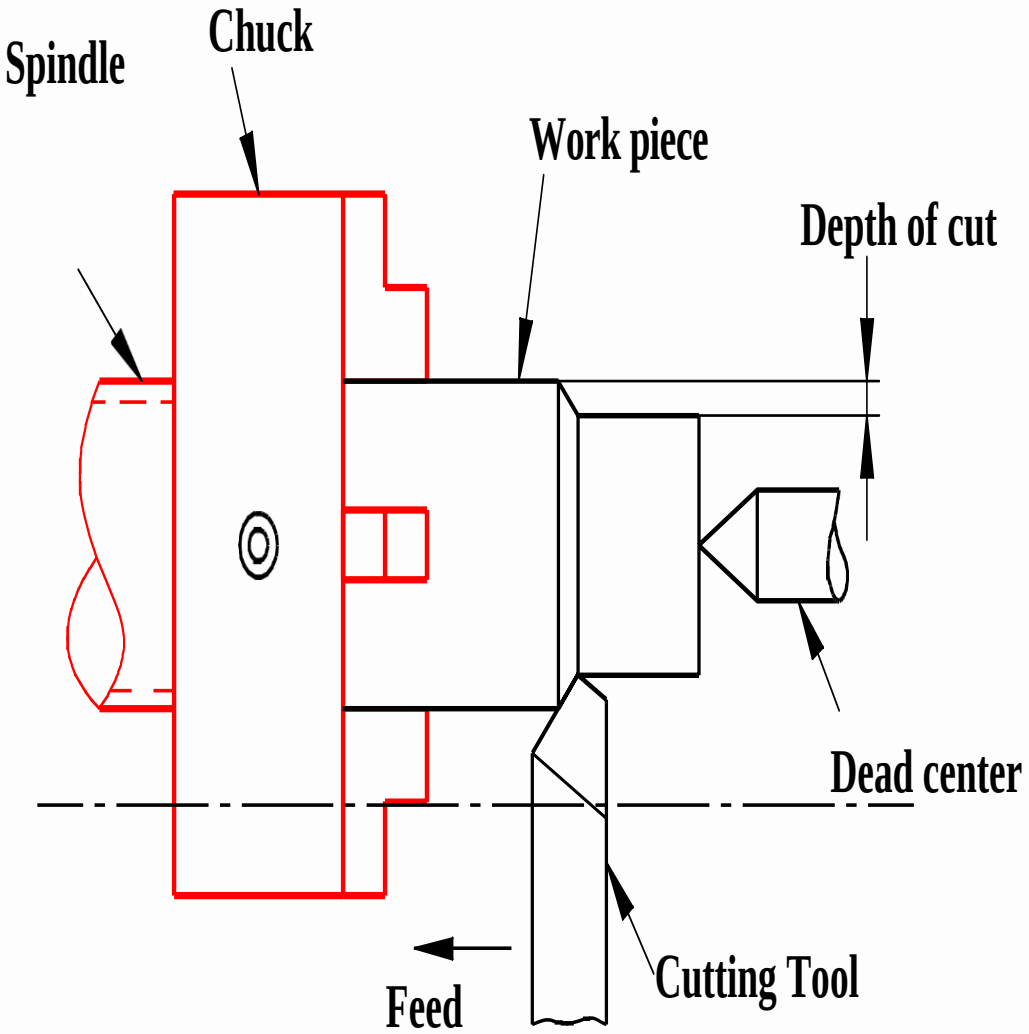
OPERATIONS OF LATHE

The most common operations performed on a lathe are:

- Turning
- Facing
- Parting
- Grooving
- Knurling
- Drilling
- Boring
- Taper turning
- Thread cutting

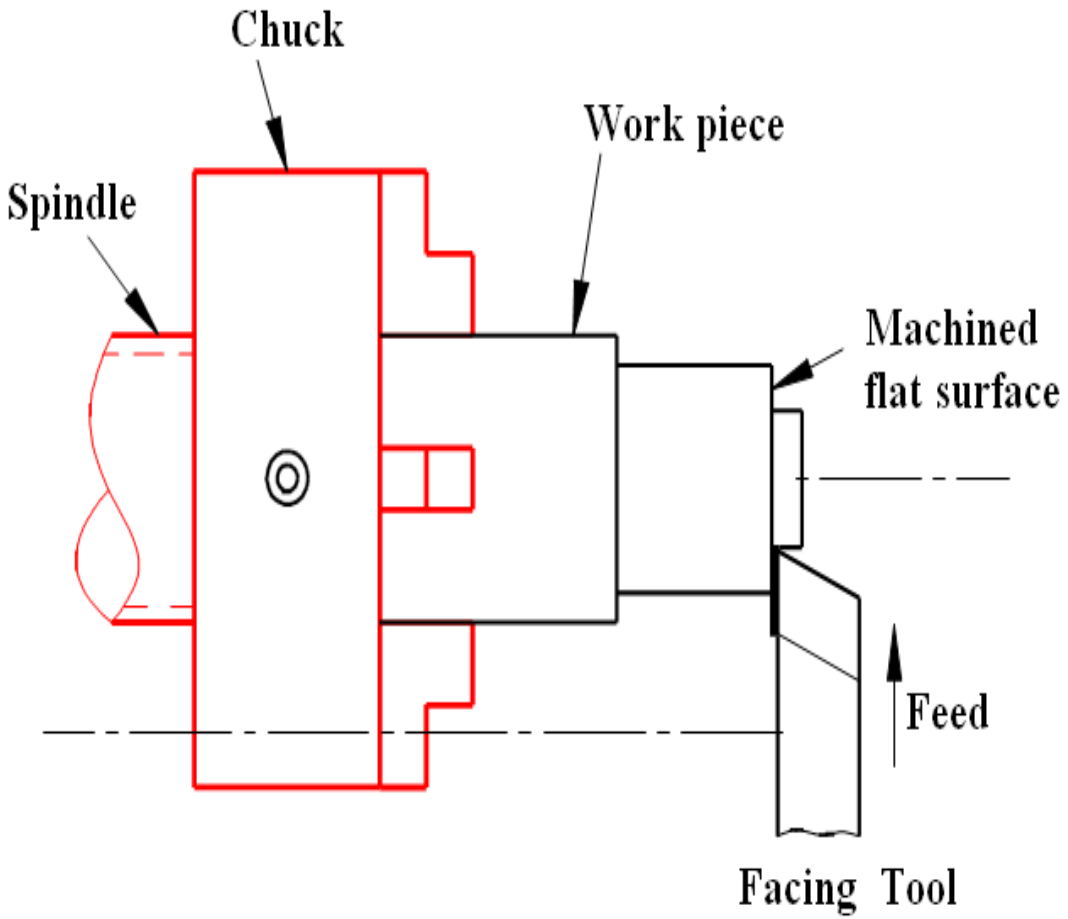
TURNING OR PLAIN TURNING:

- Plain turning is a lathe operation to remove excess material from the work piece & to produce cylindrical surface.
- Using the cross slide the cutting tool is first adjusted for desired depth of cut.
- As the work piece revolves, the tool is fed against the work-piece in a direction parallel to the axis of the spindle.



FACING:

- Facing is a lathe operation to produce a flat surface normal to the axis of rotation of the spindle.
- During facing, the carriage is locked to the lathe bed to prevent its movement.
- Using cross slide the cutting tool is fed at right angle to the axis of the work piece.



KNURLING :

- Knurling is the process of embossing a diamond shaped pattern on the surface of the work piece by the use of revolving hardened steel wheels pressed against the work.
- Knurling is done to provide grip on handles, screw heads and other cylindrical parts to be gripped by hand.



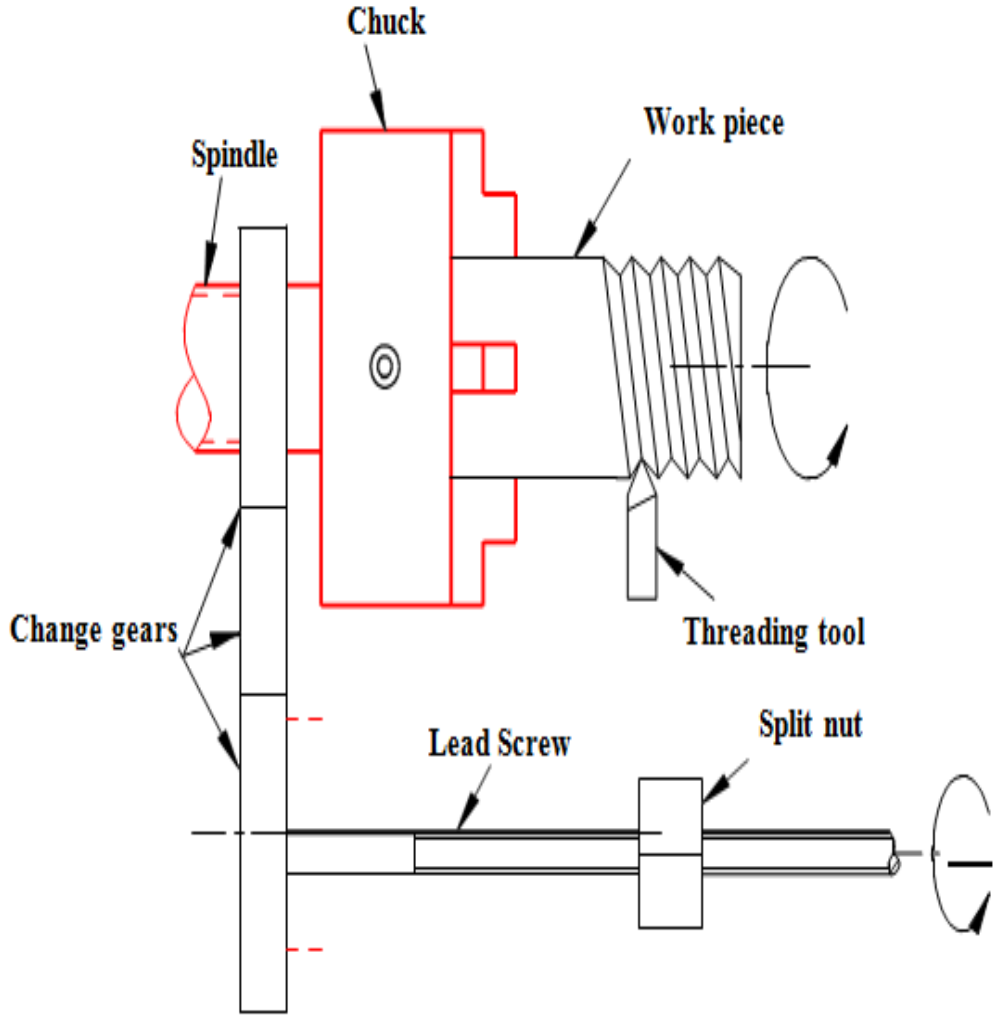
Knurling Tool



Knurled surface

THREAD CUTTING :

- The principle of thread cutting on a lathe is to produce a helical groove on a revolving cylindrical surface by feeding the tool longitudinally as shown in fig.
- The change gear calculations are made for the desired pitch of the screw and the split nut is closed.
- This engages the carriage with the lead screw and the threading tool is fed into the work using both cross feed & longitudinal feed.

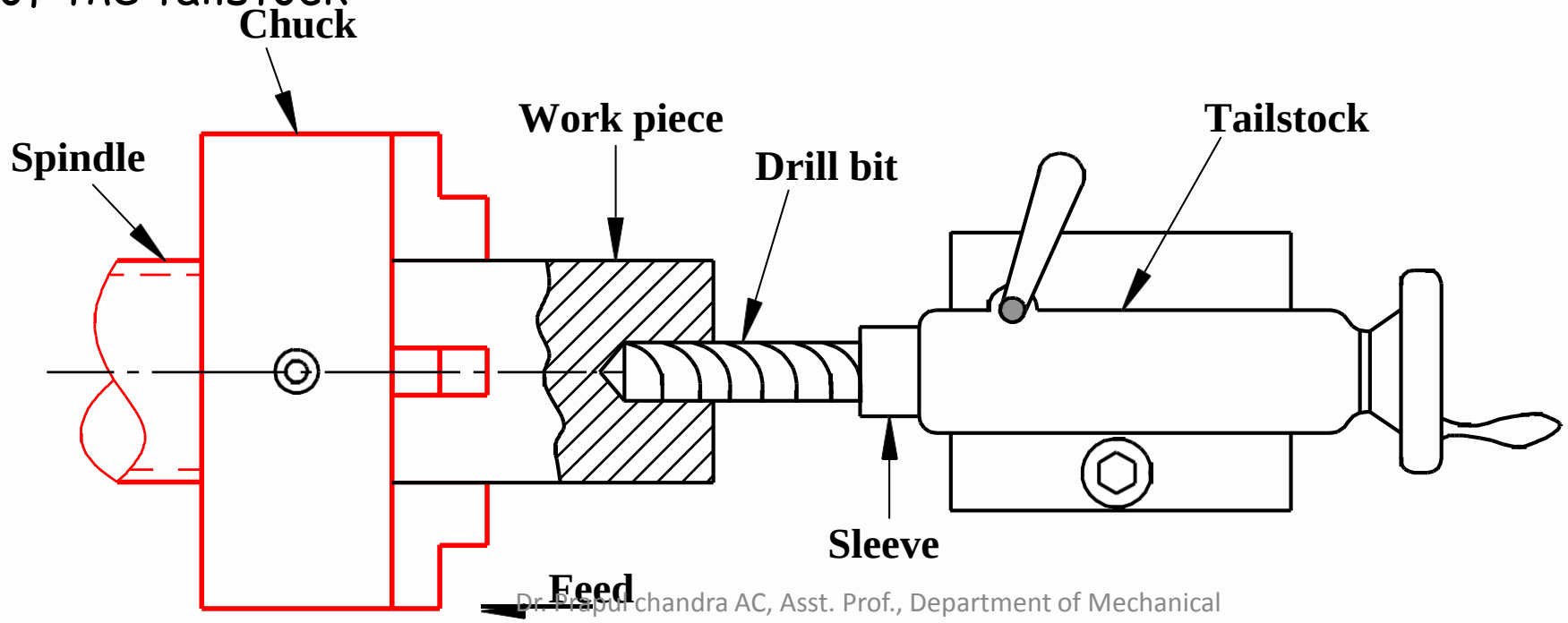


DRILLING :

- Drilling is an operation of producing cylindrical hole by means of a cutting tool called drill.
- A drill it is placed in the tapered hole of a tailstock sleeve and is fed into the work by rotating the hand wheel of the tailstock



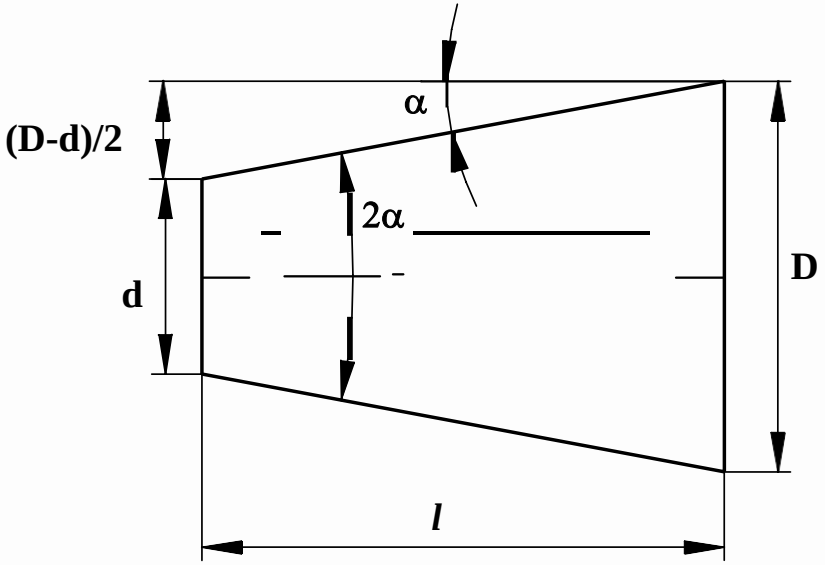
TWIST DRILL



TAPER TURNING :

- Taper turning is an operation on a lathe to produce conical surface on work pieces.
- Taper on a lathe is the difference between the diameters over a length.
- If 'D' is the larger diameter & 'd' the smaller diameter, l is the length between the two diameters, then

Taper (or) Conicity = (D-d)/ l



$$\text{Half angle of taper } \alpha = \tan^{-1}\left(\frac{D-d}{2l}\right)$$

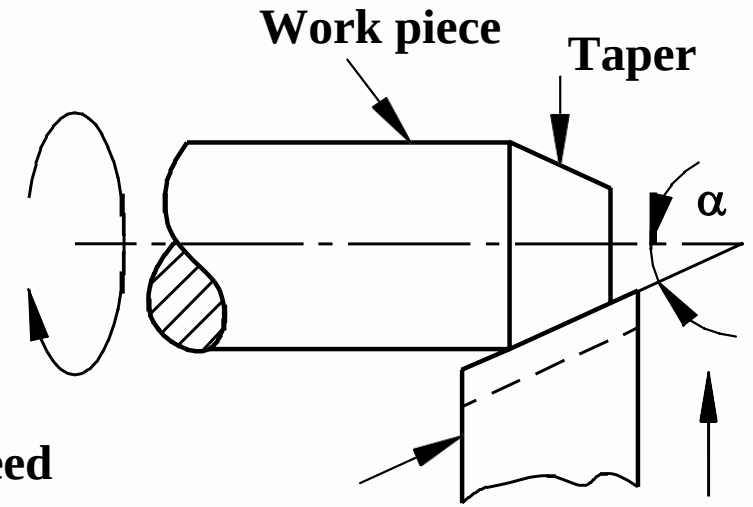
Taper Turning Methods

A taper may be turned by any one of the following methods :

- **By form tool**
- **By swiveling the compound rest**
- **By Tailstock off set method**
- **By using Taper turning attachment**

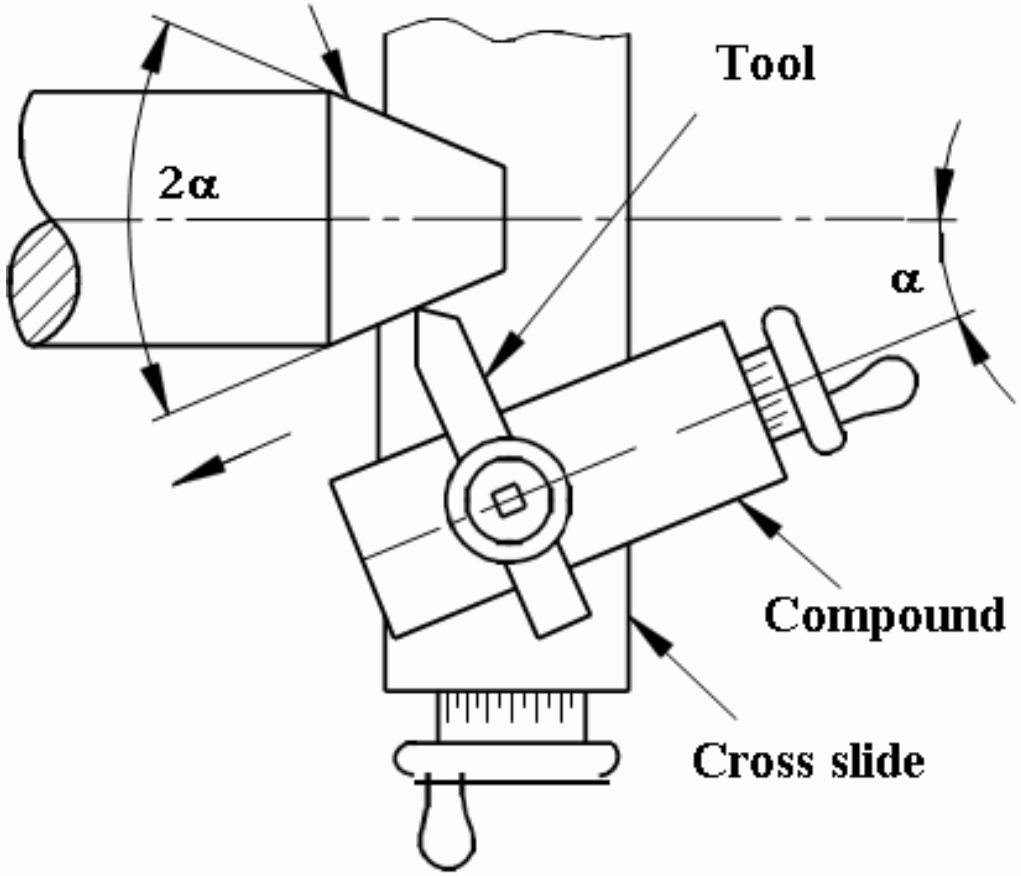
Taper Turning using form tool :

- Here a form tool which has a straight edge set at the desired angle is used.
- The angle between the cutting edge and the axis of the spindle equals half taper angle.
- This method is limited to short tapers & is mainly used for chamfering.



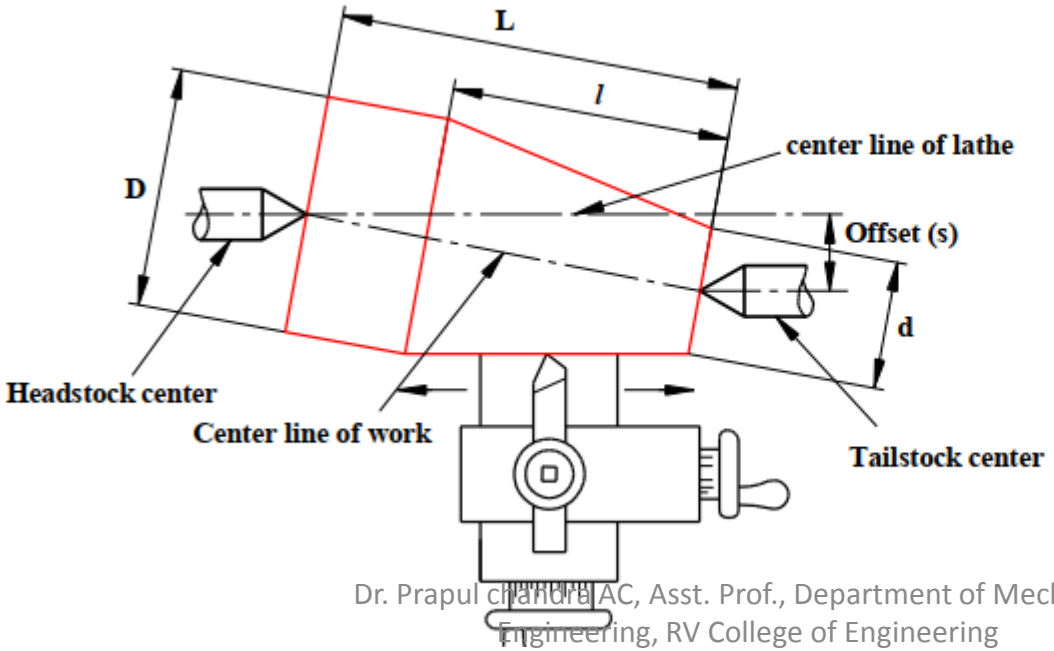
Taper Turning by swiveling compound rest :

- The compound rest may be swiveled & clamped at any desired half angle of taper on a circular base graduated in degrees.
- The tool is fed by rotating the compound slide screw.
- This is also limited to short tapers as the movement of the compound slide screw is limited.



Taper Turning by Tailstock set over (offset) :

- In this method, the center line of the work piece may be offset from the axis of the lathe and the tool is fed parallel to the axis of the lathe.
- This is achieved by moving the tailstock either towards or away from the operator by means of a set over screw. Used for small tapers on long jobs.



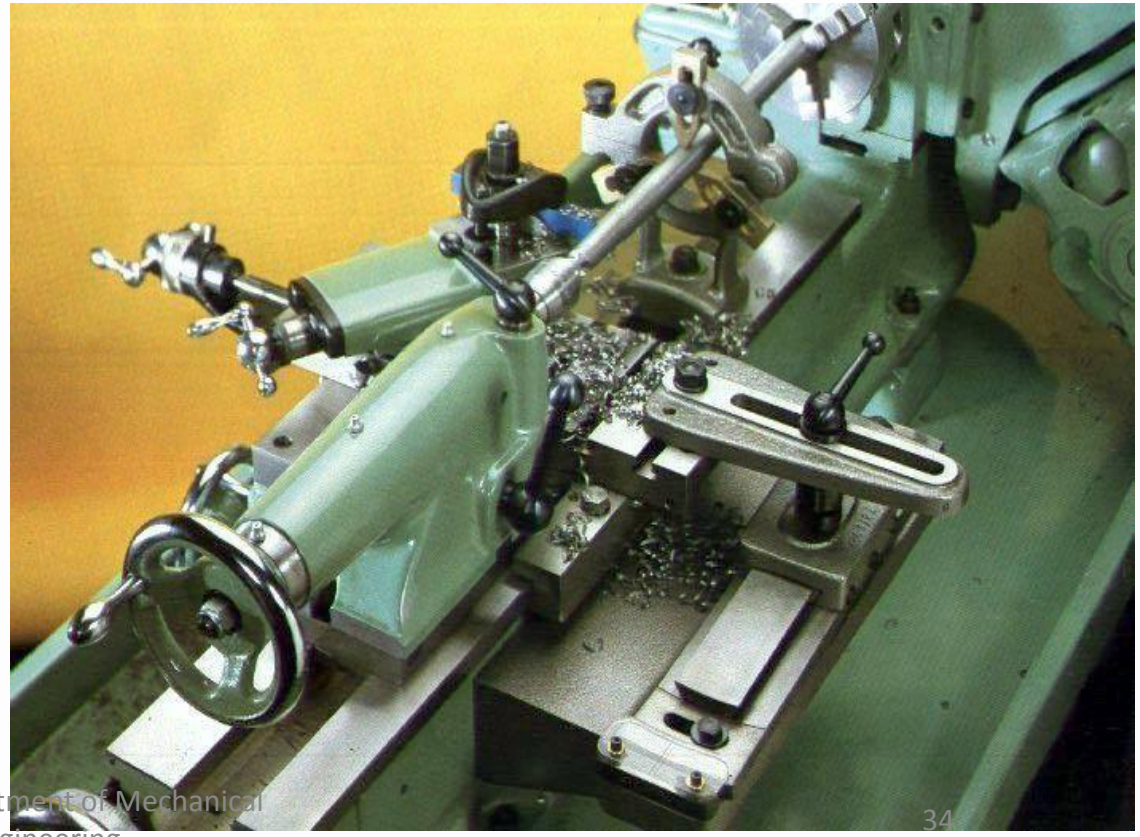
Tailstock set over

$$s = \left(\frac{D-d}{2l}\right) \times L$$

where L = length of work

Taper Turning by using attachment :

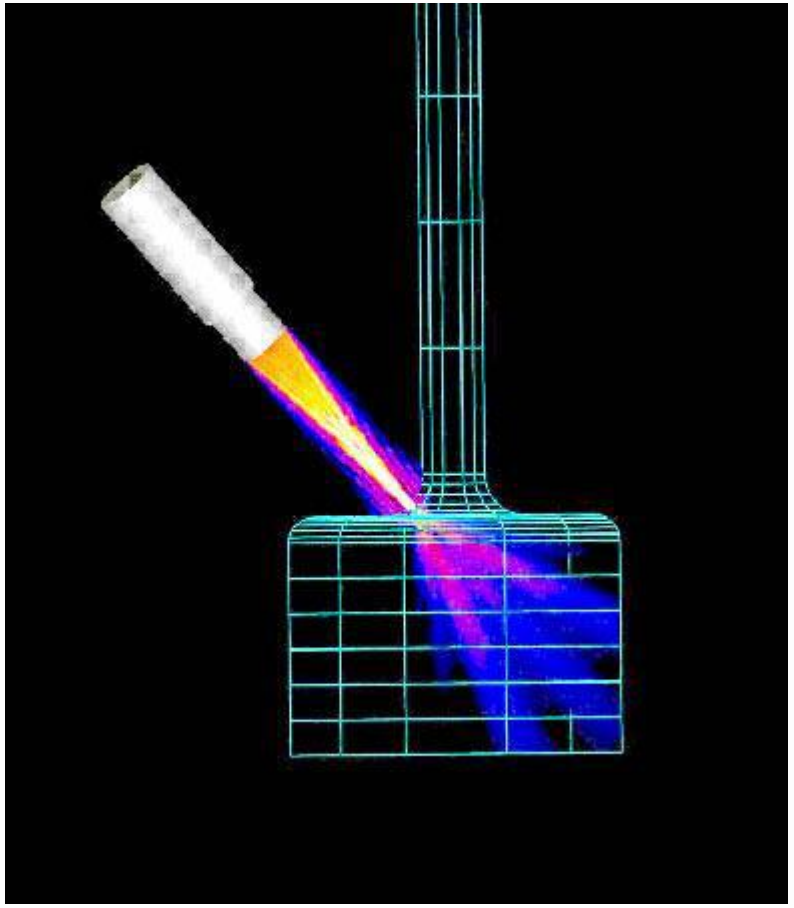
- A taper turning attachment is a fixed casting attached to the back of the lathe bed.
- It consists of a guide bar which may be set at any desired angle of taper.
- A sliding block moves over the guide bar thus moving the cross slide & hence the tool parallel to the guide bar.
- Used for long tapers with small included angles.



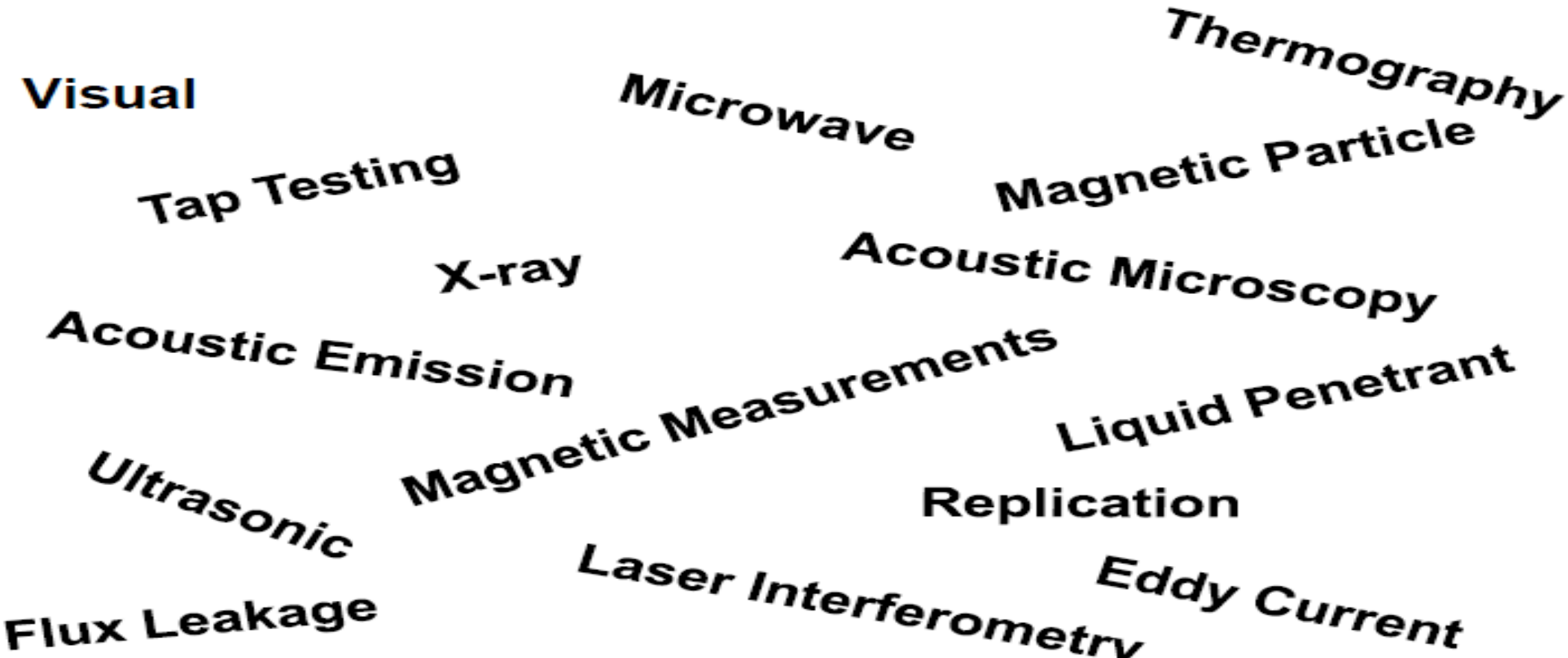
Non-destructive Testing

The use of noninvasive techniques to determine the integrity of a material, component or structure or quantitatively measure some characteristic of an object .

i.e. Inspect or measure without doing harm.



Methods of NDT



What are Some Uses of NDE Methods?

- Flaw Detection and Evaluation
- Leak Detection
- Location Determination
- Dimensional Measurements
- Structure and Microstructure Characterization
- Estimation of Mechanical and Physical Properties
- Stress (Strain) and Dynamic Response Measurements
- Material Sorting and Chemical Composition Determination



Fluorescent penetrant indication

When are NDE Methods Used?

There are NDE application at almost any stage in the production or life cycle of a component.

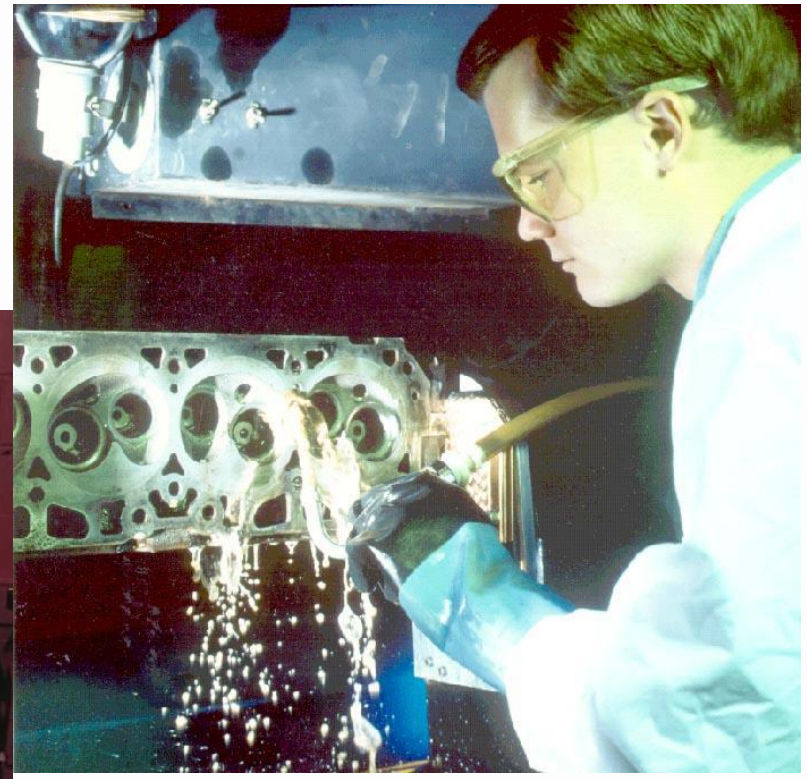
- To assist in product development
- To screen or sort incoming materials
- To monitor, improve or control manufacturing processes
- To verify proper processing **such** as heat treating
- To verify proper assembly
- To inspect for in-service damage

Most Common NDT Methods

- Visual
- Liquid Penetrant
- Magnetic
- Ultrasonic
- Eddy Current
- X-ray







Visual Inspection

- Most basic and common inspection method.
- Tools include fiberscopes, borescopes, magnifying glasses and mirrors.
- Portable video inspection unit with zoom allows inspection of large tanks and vessels, railroad tank cars, sewer lines.
- Robotic crawlers permit observation in hazardous or tight areas, such as air ducts, reactors, pipelines.







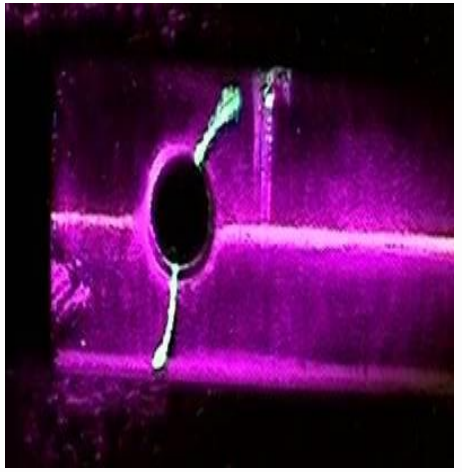
Liquid Penetrant Inspection

- A liquid with high surface wetting characteristics is applied to the surface of the part and allowed time to seep into surface breaking defects.
- The excess liquid is removed from the surface of the part.
- A developer (powder) is applied to pull the trapped penetrant out the defect and spread it on the surface where it can be seen.
- Visual inspection is the final step in the process. The penetrant used is often loaded with a fluorescent dye and the inspection is done under UV light to increase test sensitivity.

https://youtu.be/xEK-c1pkTUI



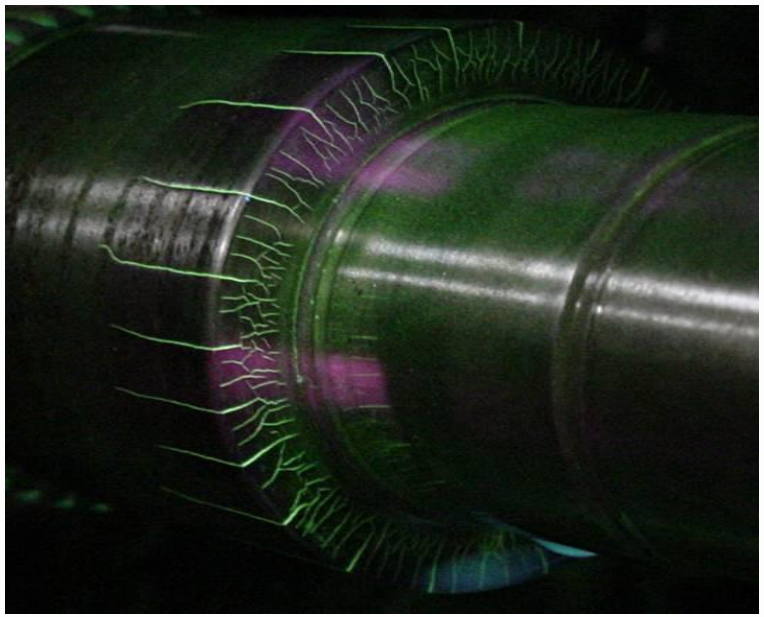


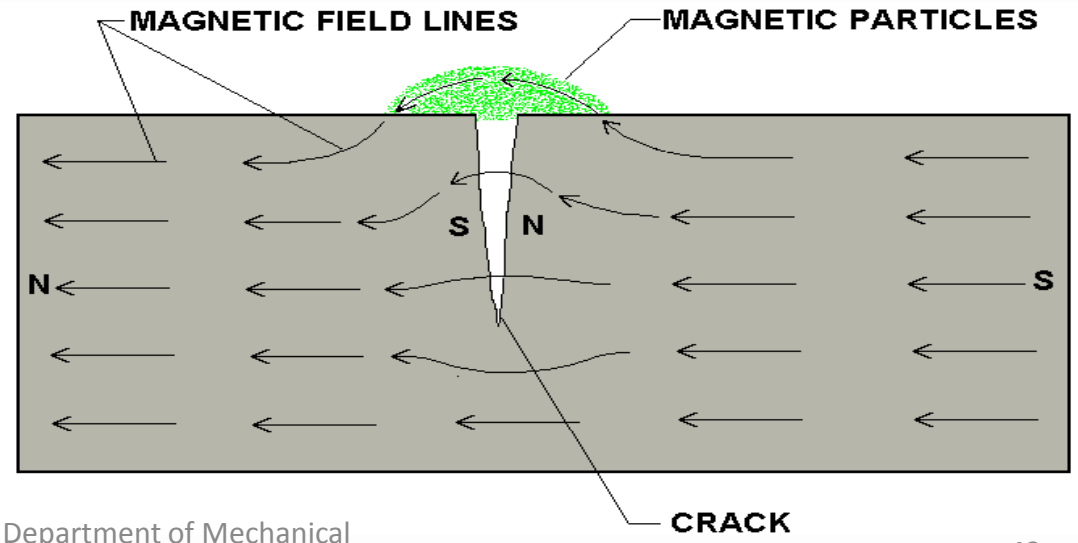




Magnetic Particle Inspection

The part is magnetized. Finely milled iron particles coated with a dye pigment are then applied to the specimen. These particles are attracted to magnetic flux leakage fields and will cluster to form an indication directly over the discontinuity. This indication can be visually detected under proper lighting conditions.

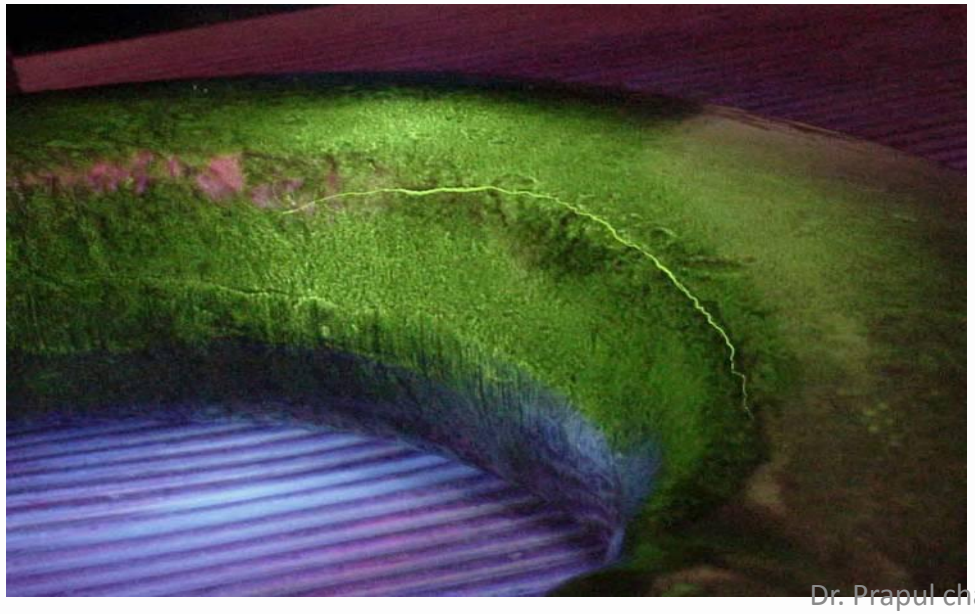


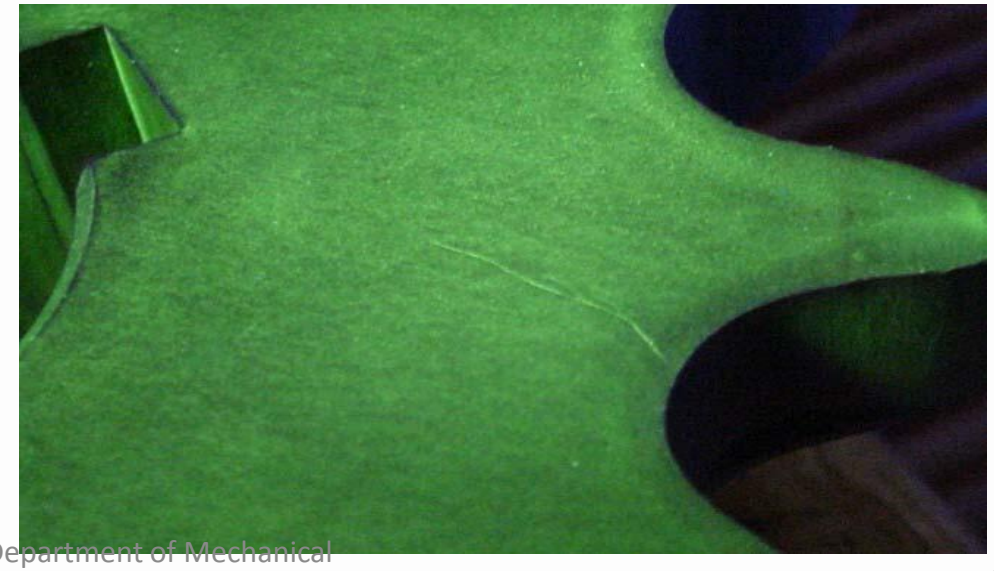


Magnetic Particle Crack Indications

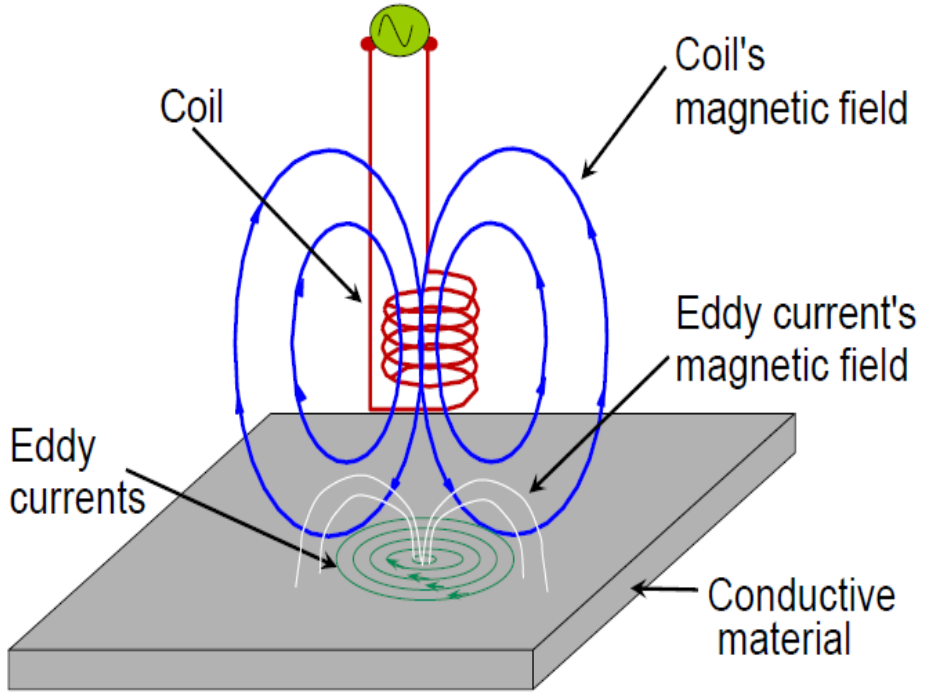








Eddy Current Testing



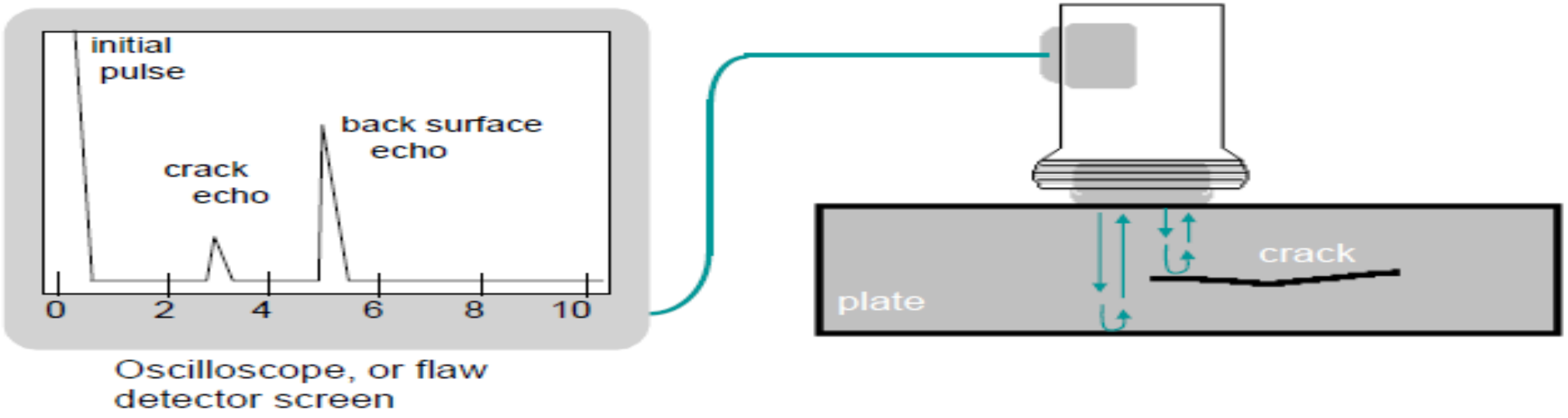
Eddy current testing is particularly well suited for detecting surface cracks but can also be used to make electrical conductivity and coating thickness measurements. Here a small surface probe is scanned over the part surface in an attempt to detect a crack.





Ultrasonic Inspection (Pulse-Echo)

High frequency sound waves are introduced into a material and they are reflected back from surfaces or flaws. Reflected sound energy is displayed versus time, and inspector can visualize a cross section of the specimen showing the depth of features that reflect sound. https://youtu.be/41i0ciqqYYM

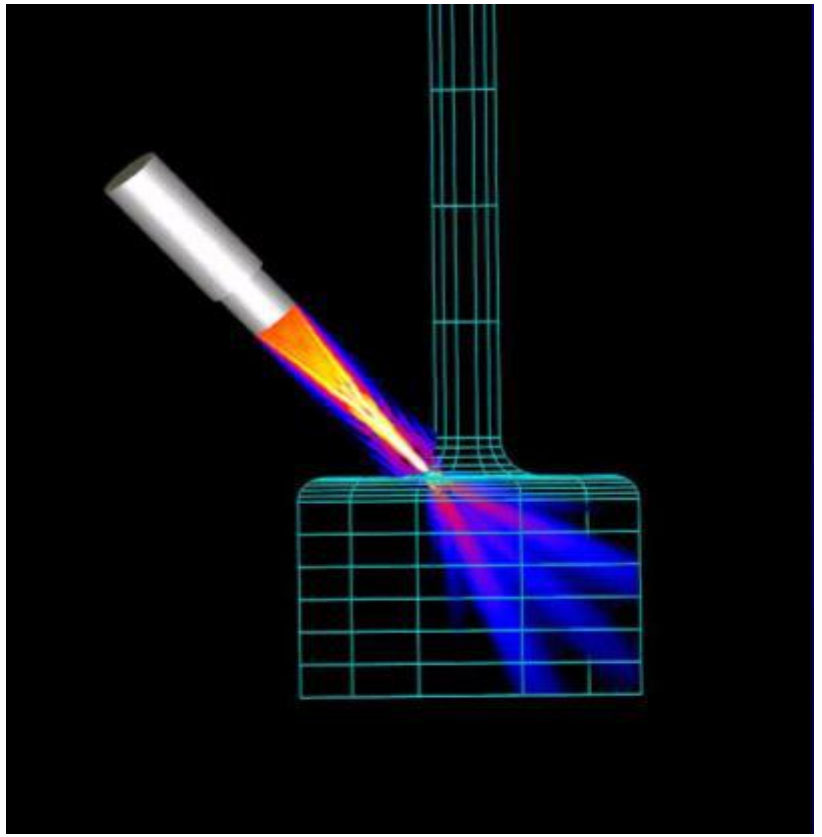


Ultrasonic Imaging

High resolution images can be produced by plotting signal strength or time-of-flight using a computer-controlled scanning system.



Gray scale image produced using the sound reflected from the front surface of the coin



Gray scale image produced using the sound reflected from the back surface of the coin (inspected from "heads" side)

Common Application of NDT

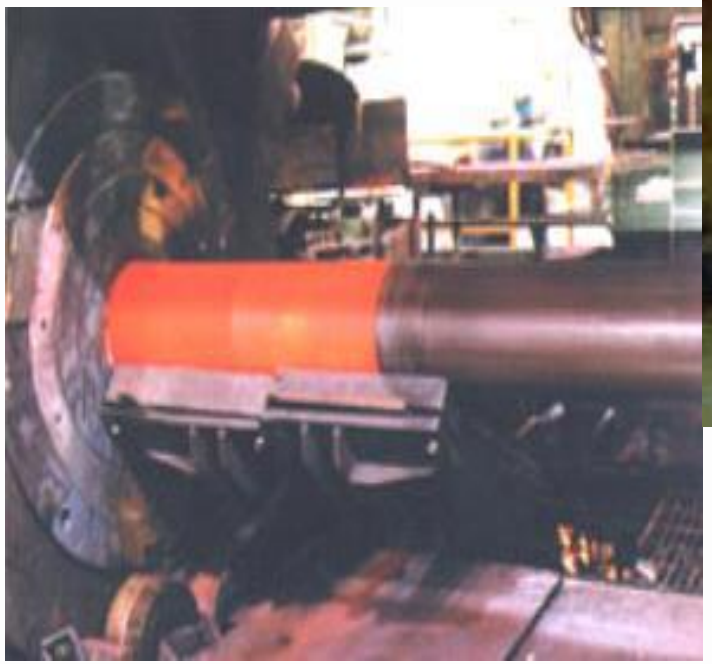
- Inspection of Raw Products
- Inspection Following Secondary Processing
- In-Services Damage Inspection

Inspection of Raw Products

- Forgings,
- Castings,
- Extrusions, etc.





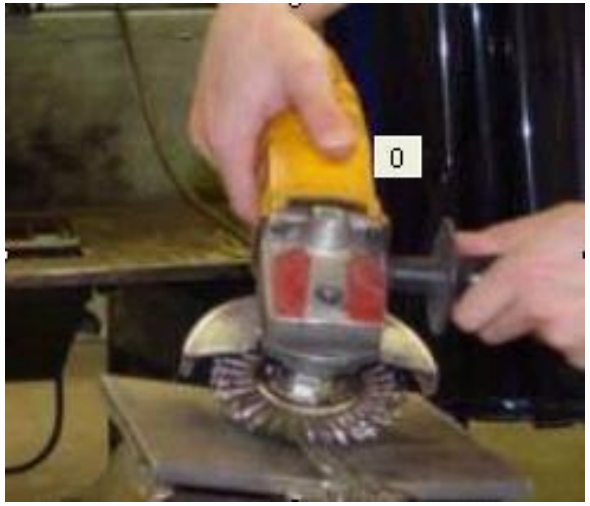


Inspection Following Secondary Processing

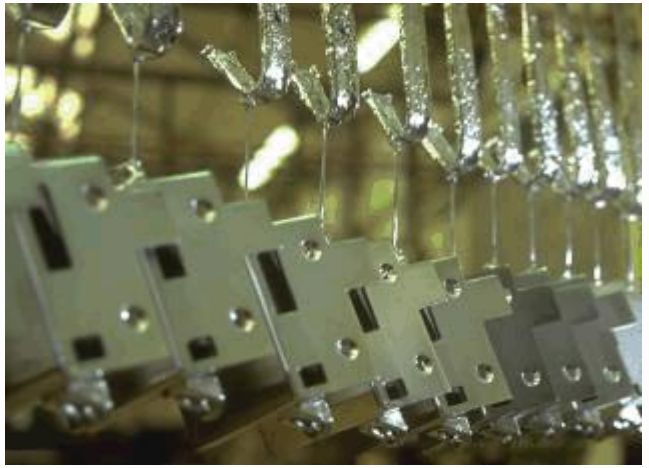
- Machining
- Welding
- Grinding
- Heat treating
- Plating











Inspection For In-Service Damage

- Cracking
- Corrosion
- Erosion/Wear
- Heat Damage





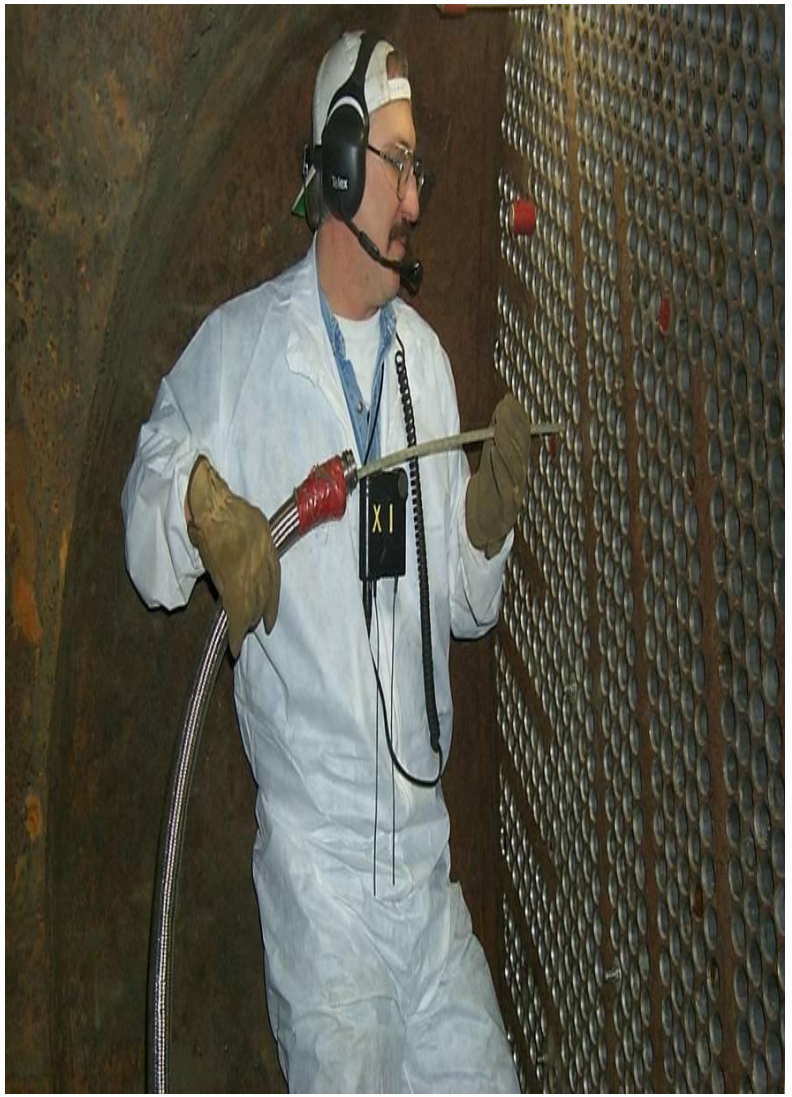




Power Plant Inspection

Periodically, power plants are shutdown for inspection. Inspectors feed eddy current probes into heat exchanger tubes to check for corrosion damage.





Wire Rope Inspection

Electromagnetic devices and visual inspections are used to find broken wires and other damage to the wire rope that is used in chairlifts, cranes and other lifting devices.







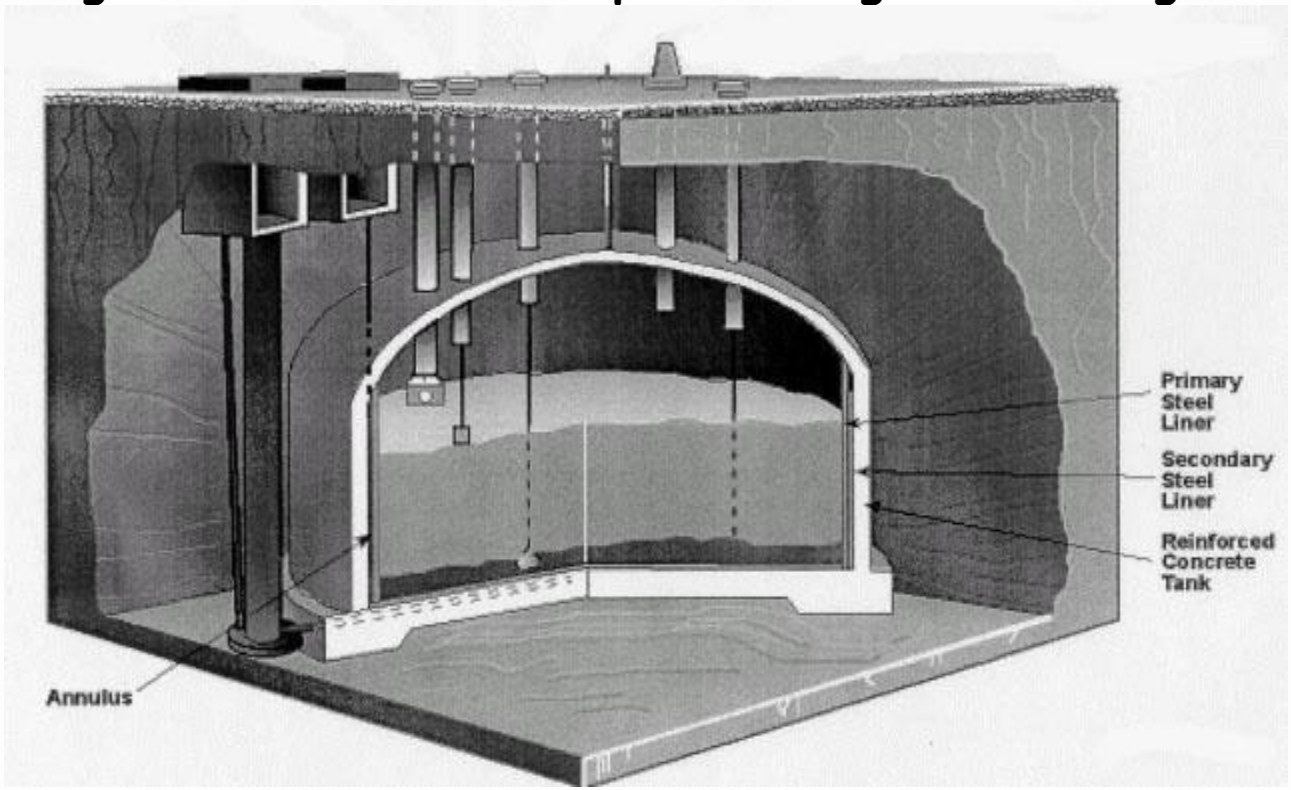


Storage Tank Inspection

Robotic crawlers use ultrasound to inspect the walls of large above ground tanks for signs of thinning due to corrosion .

Cameras on long articulating arms are used to inspect underground storage tanks for damage.



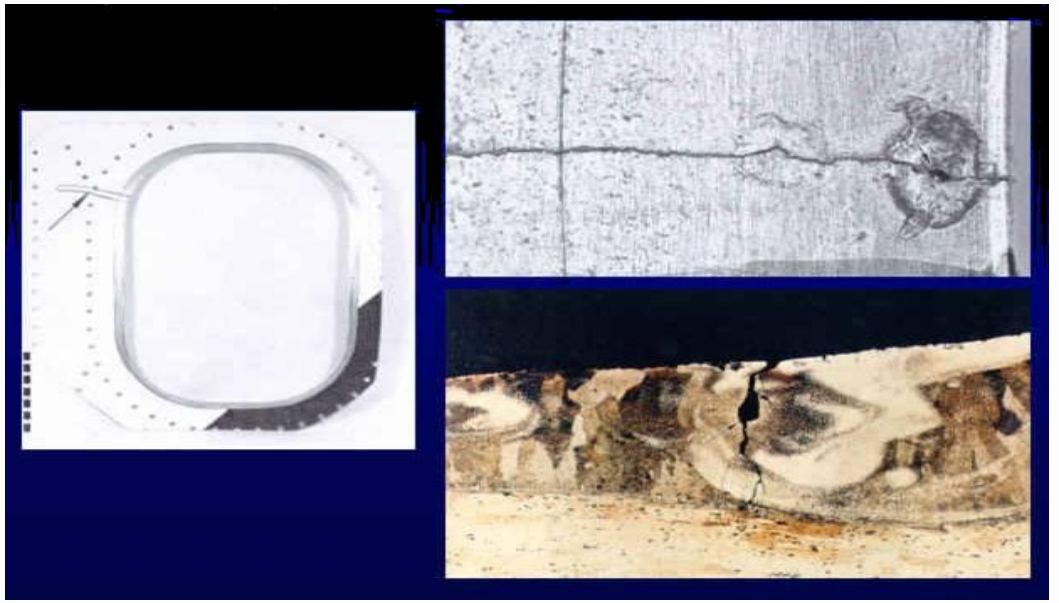




Aircraft Inspection

- Non-destructive testing is used extensively during the manufacturing of aircraft.
- NDT is also used to find cracks and corrosion damage during operation of the aircraft.
- A fatigue crack that started at the site of a lightning strike is shown below.



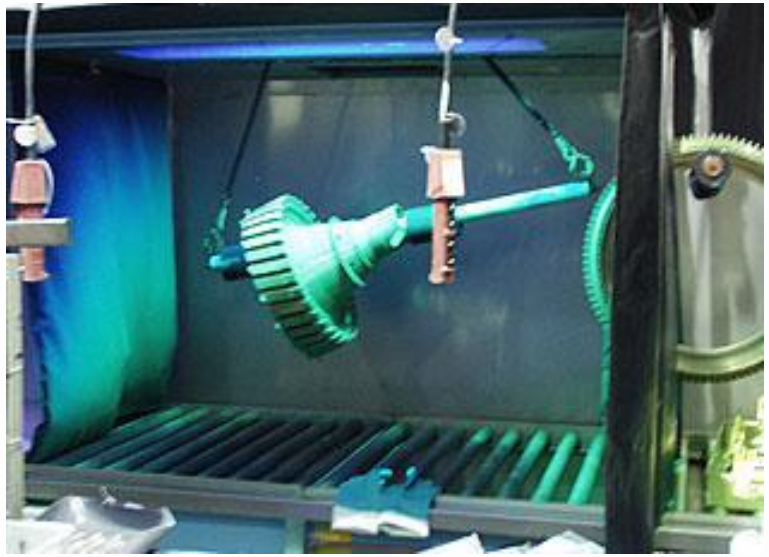




Jet Engine Inspection

- Aircraft engines are overhauled after being in service for a period of time.
- They are completely disassembled, cleaned, inspected and then reassembled.
- Fluorescent penetrant inspection is used to check many of the parts for cracking.







JOINING PROCESSES

WELDING

- Welding is a process of joining two pieces of metal by the application of heat with or without the application of pressure and filler material.
- Welding produces a permanent joint.
- Welding is used in joining pressure vessels, tanks, bridges, railways, machine frames & brackets, building the body of automobiles, aircrafts & ships etc.

Welding process may be broadly classified as;

1. Plastic welding or Pressure welding

2. Fusion welding or Non Pressure welding

1. Plastic welding or Pressure welding

In this type of welding, the metal pieces to be joined are heated to a plastic state and then joined together by the application of pressure without the addition of filler material.

Ex: Forge welding, Resistance welding.

1. Plastic welding or Pressure welding

In this type of welding, the metal pieces to be joined are heated to a plastic state and then joined together by the application of pressure without the addition of filler material.

Ex: Forge welding, Resistance welding.

2. Fusion welding or Non Pressure welding

In this type of welding, the metal pieces to be joined are heated to molten state and allowed to solidify without the application of pressure. A filler material is used during the welding process.

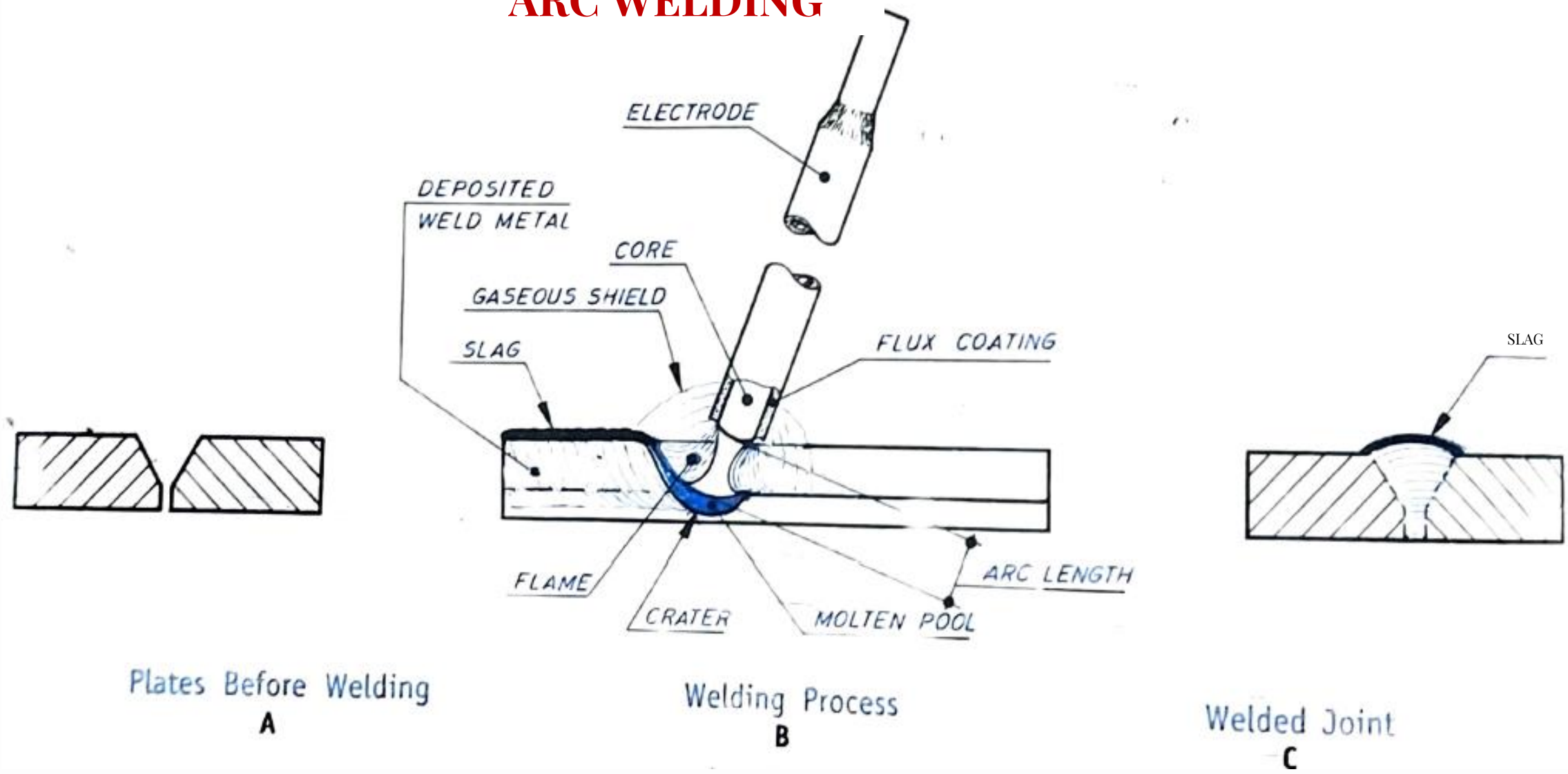
Ex: Arc welding, Gas welding.

ARC WELDING

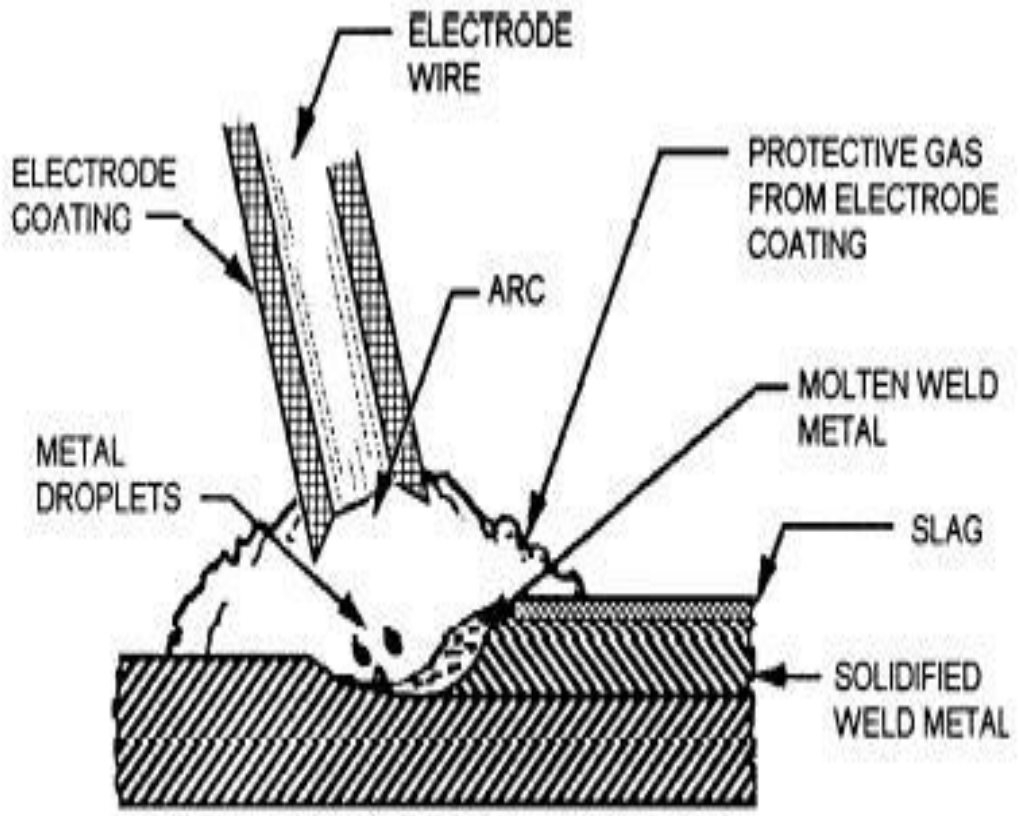
- The principle of arc welding is that, when two conductors of an electric circuit are touched together momentarily and then instantaneously separated slightly, an electric arc is formed.
- A high heat density is produced throughout the length of the arc at a temperature of 5000 to 6000 Centigrade.
- In arc welding, usually the parts to be welded are wired as one pole of the circuit, and the electrode held by the operator forms the other pole.
- When the arc is produced , the intense heat quickly melts the workpiece metal which is directly under the arc, forming a small molten metal pool.

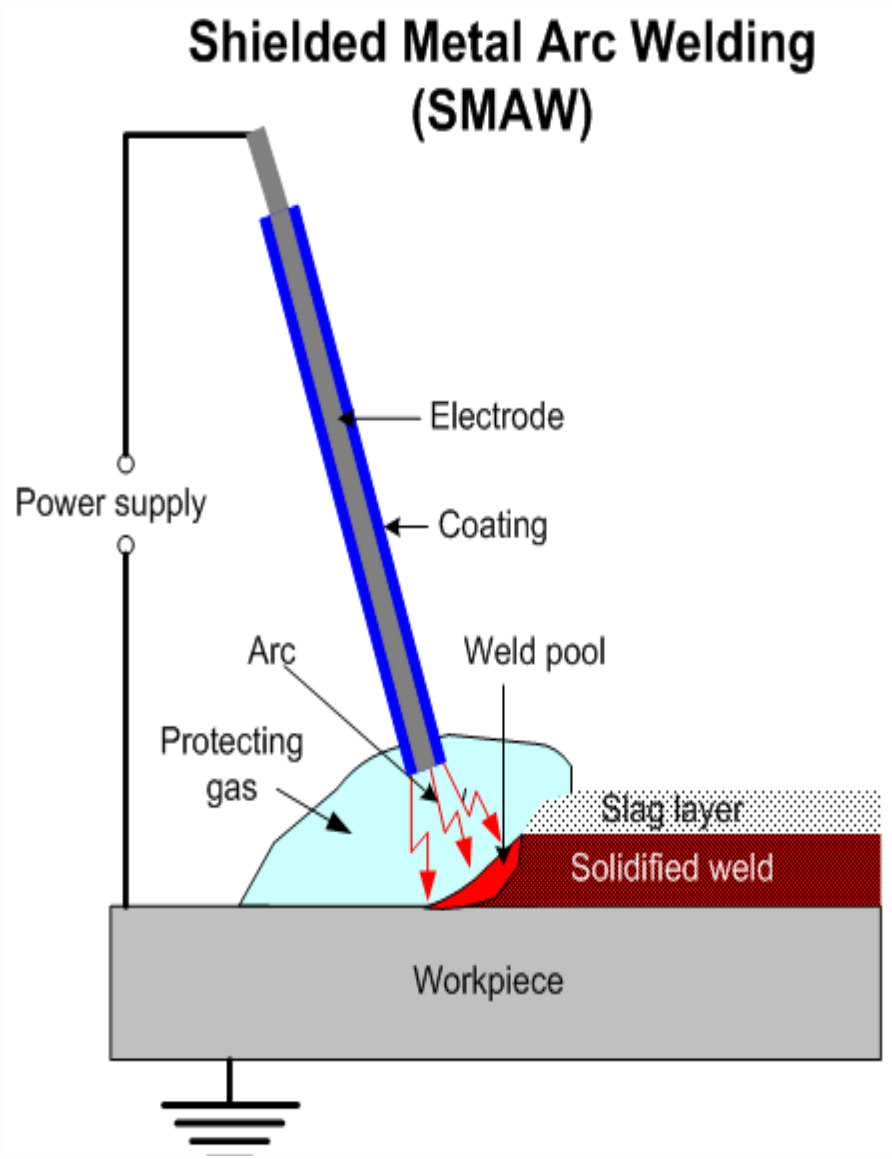
- At the same time the tip of the electrode at the arc also melts, and this molten metal of the electrode is carried over by the arc to the molten metal pool of the workpiece.
- The molten metal in the pool is agitated by the action of the arc, thoroughly mixing the base and the filler metal. A solid joint will be formed when the molten metal cools and solidifies.
- The flux coating over the electrode produces an inert gaseous shield surrounding the arc and protects the molten metal from oxidizing by coming in contact with the atmosphere.

ARC WELDING


ELECTRODE
DEPOSITED WELD METAL
CORE
GASEOUS SHIELD
SLAG
FLUX COATING
FLAME
CRATER
MOLTEN POOL
ARC LENGTH
SLAG
Plates Before Welding
A
Welding Process
B
Welded Joint
C

ARC Welding


ELECTRODE WIRE
ELECTRODE COATING
PROTECTIVE GAS FROM ELECTRODE COATING
ARC
MOLTEN WELD METAL
METAL DROPLETS
SLAG
SOLIDIFIED WELD METAL


Shielded Metal Arc Welding (SMAW)
Power supply
Electrode
Coating
Arc
Weld pool
Protecting gas
Slag layer
Solidified weld
Workpiece

Arc Welding Machine

- The function of a welding machine is to generate a low voltage (10 to 50 V) and high current. (50 A to 300 A).
- The current may be alternating current or direct current & the polarity of the electrode may be positive or negative, depending on the type of electrode and the metals to be welded.
- In A.C arc welding, a ***step down transformer is used to step down the voltage from 220/440 V to 80 to 100 V & a current of 100 to 400 A.***
- In D.C welding, the ***workpiece is connected to the +ve pole of a D.C generator and the electrode to –ve pole.***

- It is called ***'straight polarity' and is used when high heat is required.***
- When less heat is required, the polarity is reversed. Because of the option of reversing the polarity, D.C welding may be used to weld many metals which require more heat to melt.
- In A.C welding, the polarity changes in every cycle.
- Also the current & the voltage acquire a value of zero twice in every cycle, and hence higher voltage is required to maintain the arc.

ARC WELDING ELECTRODES



Welding Machine



Arc welding Electrodes

- The electrode used in arc welding is of consumable type & it is coated with flux.
- The flux coating is usually made of *chalk (lime), Ferro manganese, cellulose, Starch, Kaolin (China clay), iron powder, etc.*
- The flux forms a slag after welding which can be removed by chipping hammer & a wire brush.
- The purpose of coating the electrodes are;

(i) Protection of molten metal from oxidation

(ii) To prevent rapid cooling of molten metal

(iii) To establish & maintain the arc.

(iv) (iv) Addition of alloying elements.

PURPOSE OF EACH INGREDIENT OF FLUX COATING

- ***China clay, mica*** etc. produce a slag which because of its light weight forms a layer on the molten metal and protects the same from atmospheric contamination.
- Ingredients, like ***cellulose, wood, starch,*** calcium carbonate etc., form a protective gas shield around the electrode end, arc and weld pool.
- Deoxidizing elements like ***Ferro manganese*** refine the molten metal.
- Alloying elements like ***Ferro alloys of manganese, molybdenum*** etc. may be added to impart suitable properties and strength to the weld metal.
- ***Iron powder*** in the coating improves arc behaviour, bead appearance; helps increase metal deposition rate and arc travel speed.

SAFETY DEVICES USED IN ARC WELDING

- The welding shield protects the eyes from infra-red & ultra -violet radiations.
- Gloves are used to protect the hands from spark & to insulate from electric shocks
- Chipping hammer & wire brush to clear the slag.
- Apron to protect the clothing from sparks & spatter
- Earthing clamp will avoid the risk of electric shock.



GAS WELDING

Gas welding is a fusion method of welding in which a strong gas flame is used to raise the temperature of the workpiece to melt them. As in arc welding, a filler material is used to fill the joint.

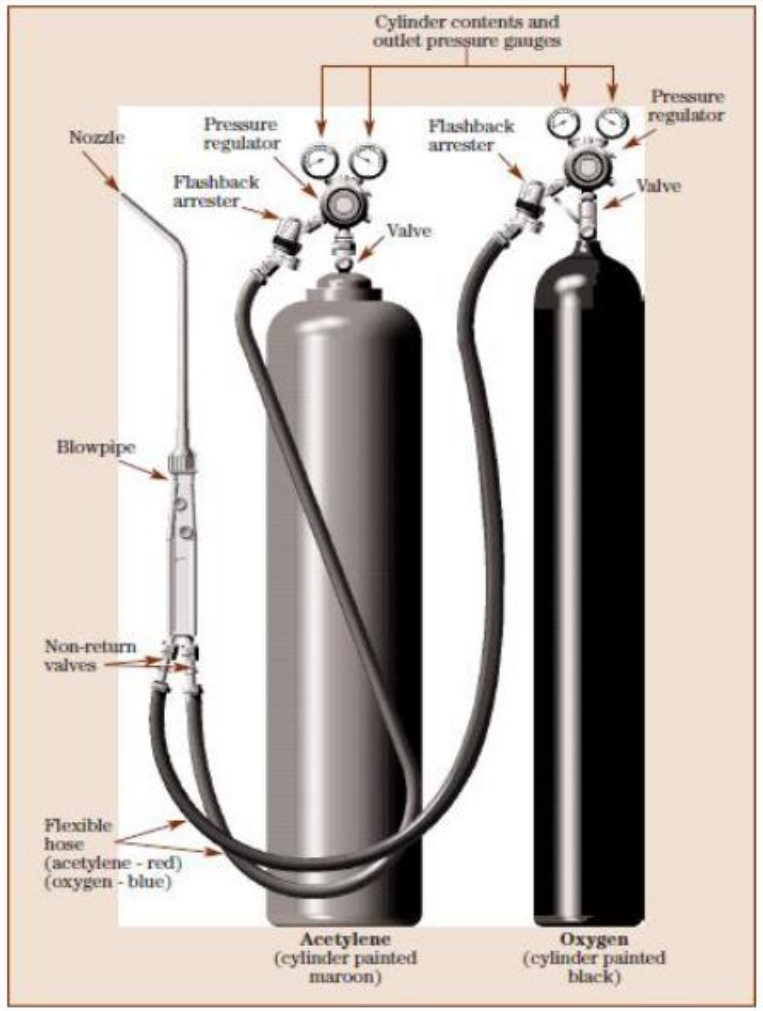
The gas combinations that can be used for heating are;

1. **Oxygen & Acetylene**
2. **Oxygen & Hydrogen**

OXY-ACETYLENE WELDING

- When right proportions of oxygen and acetylene are mixed in the welding torch and when ignited, the flame produced at the nozzle tip is called as the *Oxy-Acetylene Flame*. This flame when used in welding becomes Oxy-acetylene Welding.
- The temperature attained by the Oxy-acetylene flame is around 3200 Centigrade and therefore has the ability to melt all commercial metals. Thus, there is a complete bonding of the joining metals that can be achieved during welding

JOINING PROCESSES

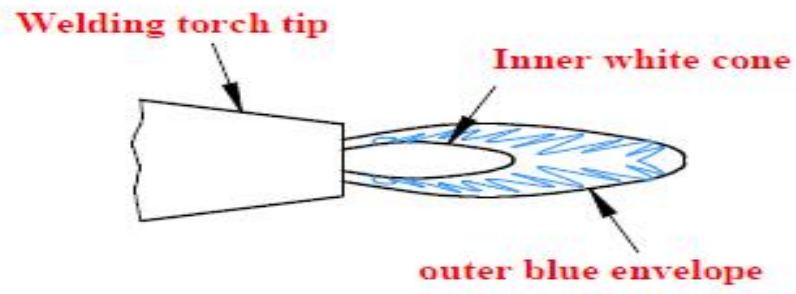


- The oxy-acetylene gas equipment consists of two large steel cylinders; one containing oxygen at high pressure and the other dissolved acetylene also at high pressure.
- It also consists of rubber tubes, pressure regulators, & blow torch.
- The oxygen and acetylene are supplied to the blow torch separately where both of them get mixed together and come out through the nozzle of the blow torch.

TYPES OF OXY-ACETYLENE FLAMES

Neutral flame

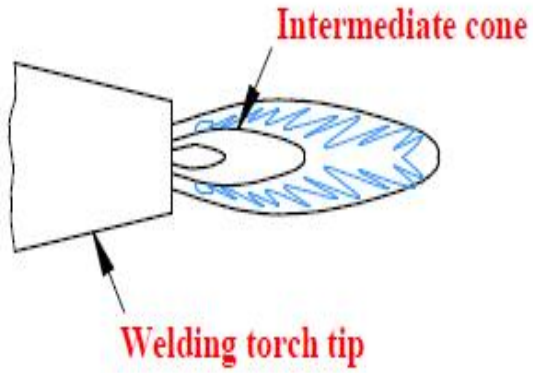
- A Neutral flame is obtained by supplying equal volumes of oxygen & acetylene. (Oxygen: Acetylene = 1: 1)
- A neutral flame consists of an inner small whitish cone surrounded by a sharply defined blue flame.
- Most of the oxy-acetylene welding is done with the neutral flame.





CARBURIZING FLAME

- A Carburizing flame or reducing flame is obtained by supplying excess acetylene. (Oxygen : Acetylene = 0.95 :1)
- It has three cones; an inner white cone, surrounded by an intermediate whitish cone & a bluish envelope flame.
- Carburizing flame is used for welding alloy steels, cast iron & aluminium to protect from oxidizable elements

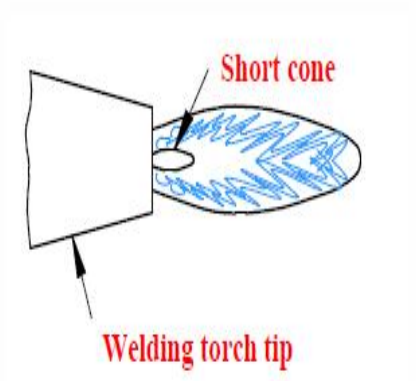


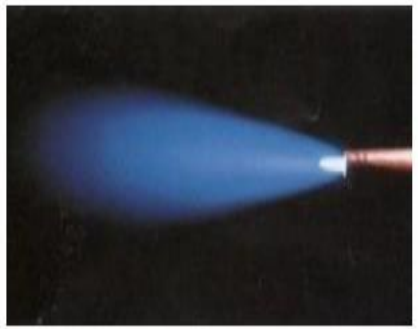


OXIDIZING FLAME:

- The Oxidizing flame is obtained when there is excess oxygen having gas ratio as high as 1.5. (Oxygen: Acetylene = 1.5 : 1)
- In appearance, it resembles a neutral flame except that the inner white cone is somewhat shorter.
- It is used for welding nickel and many non ferrous materials.
- This is used for oxy-acetylene cutting and is not suitable for welding, since the weld metal will be oxidized

.





Defects in welding Common weld defects include:

Lack of fusion: It results from too little heat input and too rapid traverse of the welding torch (gas or electric)

Porosity: This occurs when gases are trapped in the solidifying weld metal.

Inclusions: These can occur when the slag covering a weld is not totally removed before another weld run.

Cracking: This can occur due to thermal shrinkage.

Undercut: This is due to excess melting of the parent metal which reduces the strength of the joint.

SOLDERING

- **Soldering is a process in which two or more metal items are joined together by melting and flowing a filler metal (solder) into the joint.**
- The filler metal will have a lower melting point than the workpiece ***(between 1500C-3500C).***
- Soldering differs from welding in that ***soldering does not involve melting the work pieces.***
- An alloy of lead & tin called '**soft solder**' is used for sheet metal work, plumbing work & electrical junctions. The melting temperature of the soft solder will be between 150 to 350 C.
- To clean the surfaces to be joined & to prevent oxidation, Zinc chloride is used as a flux.
- A soldering iron is used to apply heat produced from an electrical source.
- An alloy of Copper, Tin and Silver known as **Hard solder** is used for stronger joints. The soldering temperatures of hard solders ranges from 600 to 900 C

Method of Soldering

- **Cleaning of Joining surfaces:** The joining surfaces are cleaned mechanically to make them free from dust, oil, scales etc and ensure that the molten filler wets the surfaces.
- **Application of flux:** The joining surfaces are coated with a flux, usually rosin or borax. This cleans the surfaces chemically and helps the solder in making a bond.
- **Tinning of the surfaces to be soldered:** Before carrying out the soldering operation, the soldering iron must be “tinned”. This is to remove a thin that forms on the copper bit.

Method of Soldering

- **Heating:** The soldering iron is then heated and the flowing molten filler metal fills the joint interface. Allow the soldered area to cool and then solidify thus making the joint.
- **Final clean Up :** After completing the soldering, and the joints are formed, clean it with steel wool or solvent to remove left –over flux. After this clean the soldering iron using a damp sponge.



Soldering Iron



Soldering Flux



Soldering wire

SOLDERING INSTRUMENTS